

graphene sheets are incorporated into composites, you could come up with a material that's many times stronger than Kevlar. The Chinese are already working on carbon-nanotube yarn for spacesuits and bulletproof vests.

Graphene composites could be produced less expensively than the current generation of carbon-nanotube composites. That opens the way for lighter, cheaper body armour as well as lighter auto bodies and airplane fuselages.

T-ray scanners. Terahertz radiation, or T-rays, is particularly well-suited for detecting hidden objects at airport security checkpoints without the health risk posed by X-rays. T-rays could also serve as the basis for medical scanning devices that come even closer to the 'Star Trek' tricorder. T-ray scanning is already being used for skin-cancer screening and tooth-cavity detection.

The fast frequencies that can be achieved using graphene circuits are the basis for chemical sensors and generators of terahertz-range light. Maybe graphene will hasten the arrival of those powerful new T-ray scanners, for better or worse.

Heat dissipation to cool electronics. Overheating in laptops and other electronic gadgets is not just an annoyance to the end user, but it's a major technological hurdle to the speed and energy efficiency of electronic products. Heating in electronic components is inevitable and, as processing speeds increase exponentially, this problem needs to be dealt with using constantly improving technology.

While heat-dissipating fans and the increasingly popular water cooling systems may do the job for now, more portable (and/or quieter) solutions are needed. Silicon—an invaluable material for its unique electronic properties—doesn't have good thermal properties, particularly at the nanometre scale. A new, promising approach to controlling the heat problem is therefore to incorporate materials with superior thermal properties into silicon computer chips, to make the heat transfer swifter and more efficient.

Graphene behaves as a strong heat conductor, which could mean lower temperatures and a concrete possibility for chip manufacturers to reach higher processing speeds with relative ease. Recently, a team of scientists from the University of California (USA) found that multiple layers of graphene show strong heat-conducting properties that can be harnessed in removing dissipated heat from electronic devices.

Further developments

Discovered only four years ago, graphene already appears in prototype transistors, memories and other devices. But for it to go from lab benches to store shelves, engineers need to devise methods to make industrial quantities of large, uniform sheets of pure, single-layer graphene.

Researchers are pursuing several processing routes. The initial graphene-making methods worked similarly to pencil writing: researchers would abrade some graphite and then search the debris with a microscope for suitable samples or separate individual flakes with sticky tape. But such mechanical exfoliation techniques are considered to be suitable only for making tiny amounts.

Many research groups claim to be able to coat entire silicon wafers with monolayer sheets of graphene cheaply but so far no one has publicly demonstrated it. Recently, the procedure was scaled up to produce as much graphene as you want by using ultrasound to break up graphite into individual layers that are dispersed in a liquid. The suspension can then be dried out on a surface, which leaves a film of overlapping pieces of graphene crystals. Whether these sheets of multiple crystals can work well enough for many applications is uncertain, however, because edge boundaries of individual flakes tend to impede the rapid flow of electrons. Bigger samples might come from chemical exfoliation.

Researchers at the Massachusetts Institute of Technology and elsewhere are looking to make graphene using

chemical vapour deposition (CVD)—an established process that could be readily integrated into microchip fabrication. In CVD, volatile chemicals react and deposit themselves on a substrate as a thin coating.

The biggest challenge in exploiting graphene's jack-of-all-trades flexibility in computing applications is getting it to perform as a true semiconductor. While graphene can be considered a semiconductor like silicon, it lacks one crucial property—the ability to act as a switch. Without this, a chip will draw electricity continuously, unable to turn off.

But engineers are making headway. Researchers at the University of Illinois have showed that nanoribbons of graphene could be cut such that they could be turned on and off.

The most likely applications for graphene will be in analogue systems, such as radar, satellite communications and imaging devices. The first graphene device from the Defense Advanced Research Projects Agency (DARPA) of the United States Department of Defense will be for specialised government communications. Researches envision graphene transistors amplifying signals between cell towers and eventually inside cell phones. Even in analogue devices, there are still hurdles to overcome—most notably the difficulty of creating large batches of graphene. Graphene research has discovered hidden interactions that will affect the way components are designed from the super-fast material.

Scientists from the Georgia Institute of Technology and the US National Institute of Standards and Technology (NIST) have determined how the orbits of electrons interact with magnetic fields applied to epitaxial graphene. Understanding such interference will be important for bilayer graphene devices that have been proposed, and other lattice-matched substrates used to support graphene and graphene devices. ●

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